

Take-Home Assignment weeks 37–38

Noah Rahbek Bigum Hansen

Exercise P5-1

The pipe is hoisted using the tongs. If the coefficient of static friction at A and B is μ_s , determine the smallest dimension b so that any pipe of inner diameter d can be lifted.

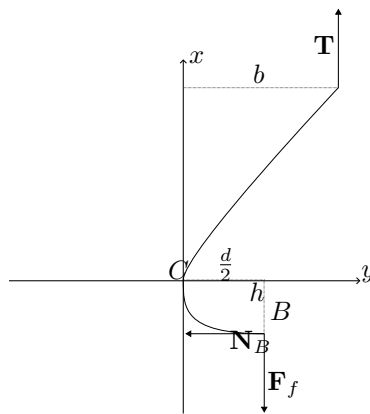
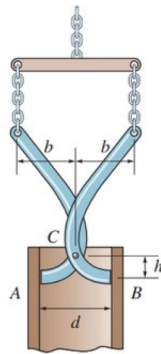


Figure 0.1: Free-body diagram of the right tong half

We consider one half of the tong, in this case the right half as shown on Figure 0.1. For static equilibrium

to occur we have the necessary and sufficient conditions:

$$\begin{aligned} F_{R_x} &= \sum F_x = 0 \\ F_{R_y} &= \sum F_y = 0 \\ M_{R_C} &= \sum M_C = 0. \end{aligned}$$

For a couple moment not to arise, the lines of actions of all the forces must intersect at some common point, P .

At point B , right at the verge of slippage, the line of action of the resultant force at B makes the friction angle, $\phi = \tan^{-1} \mu_s$ below the horizontal. Using the equation of a line we get:

$$y + h = \mu_s \left(x - \frac{d}{2} \right) \implies y = \mu_s \left(x - \frac{d}{2} \right) - h.$$

The line of action at point B will lie along this line.

At the top the tong is attached to a vertical chain, such that the force that is transmitted to the tong at the top is vertical, with a line of action of $x = b$.

As mentioned above, all the lines of actions of the forces must intersect at a common point P . This means that the two lines defined above must intersect at this point P . This intersection will be:

$$P = \left(b, \mu_s \left(b - \frac{d}{2} \right) - h \right).$$

The pin force at C will thus act along the line CP . If this point P is below point C a counter clockwise torque is induced and thus the jaws of the tong will tend to open up. Therefore, for the jaws to wedge and hold, the concurrency point P must be at or above C as:

$$\begin{aligned} y_p &\geq 0 \\ \mu_s \left(b - \frac{d}{2} \right) - h &\geq 0 \\ \implies \mu_s \left(b_{\min} - \frac{d}{2} \right) - h &= 0 \\ b_{\min} &= \frac{h}{\mu_s} + \frac{d}{2}. \end{aligned}$$

Therefore the smallest dimension of b such that the pipe can be lifted is $b_{\min} = \frac{h}{\mu_s} + \frac{d}{2}$.